

Architectural Pattern Selection – Self-Service Coffee Kiosk

Selected Architecture: Layered Architecture

For the self-service coffee kiosk, **Layered Architecture** is selected as the most appropriate architectural style. The system is divided into three main layers: Presentation Layer, Business Layer, and Data Layer. This separation makes the system easier to understand, maintain, test, and modify.

The Presentation Layer contains the User Interface Component. It allows customers to interact with the kiosk, select menu items, customize orders, and make payments.

The Business Layer contains the Order Manager, Payment Service, and Receipt Printer components. The Order Manager handles order processing and coordinates with the Payment Service for payment processing and the Receipt Printer for generating receipts. This keeps the main business logic separate from the user interface.

The Data Layer contains the Database Component. It stores and retrieves menu information, pricing, and other required order-related data. The Order Manager communicates with the database through the Database Interface.

The component diagram shows four main interfaces: Order Interface, Payment Interface, Receipt Interface, and Database Interface. The provided and required interfaces show how services are supplied and consumed between components.

Layered Architecture is suitable because changes in one layer can be made with minimal impact on other layers. For example, the user interface can be modified without significantly changing the payment or database components. It also provides clear separation of responsibilities and makes component interactions easier to understand.